

Experiment No : 03

Experiment Name : Study of Zener Diode as a voltage regulator

Objectives:

- To implement zener diode as a voltage regulator
- To observe the output waveforms of the voltage regulator circuit

Theory:

A Zener Diode is a special type of semiconductor that conducts in reverse once the voltage reaches its breakdown level, called the Zener Voltage (V_Z). Below this voltage, it stays off like a normal diode. When V_Z is reached, it safely enters breakdown and allows controlled reverse current, making it useful for regulation.

As a voltage regulator, the Zener Diode keeps the output voltage stable even if the input or load changes. In reverse bias, it remains off when $V_{in} < V_Z$, so the full voltage appears across the load. When $V_{in} > V_Z$, it enters breakdown and holds the load voltage nearly constant at V_Z . A series resistor drops the extra voltage and limits current to protect the diode.

Required Apparatus:

- Zener Diode (1 piece)
- Resistor ($1\text{ k}\Omega$, $10\text{ k}\Omega$; 1 piece each)
- DC supply (0-15 V)
- Project Board
- Multimeter
- Connecting wires

Circuit Diagram:

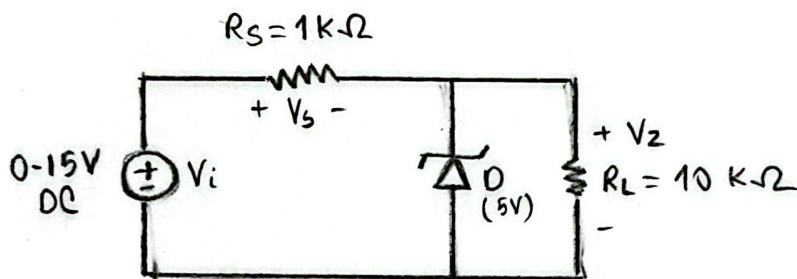


Fig: Circuit Diagram for implementing Zener Diode as a voltage regulator

Procedure:

- The circuit was connected according to the above circuit diagram
- The input voltage (V_{in}) was gradually increased and the corresponding voltage across the load (V_L) was recorded
- Step-(ii) was repeated until the voltage across the zener diode exceeds its zener voltage.
- It was observed whether the load voltage remained constant when the voltage across the zener diode exceeded its zener voltage.

Data Table:

Supply Voltage, V_i (V)	Voltage across zener diode, V_z (V)	Voltage across series resistance, V_s (V)
5.01	4.46	0.53
7.03	4.99	2.02
10.28	5.05	5.22
12.03	5.06	6.96
15.00	5.07	9.92

Discussion and Conclusion:

This experiment confirmed that the Zener Diode kept the load voltage nearly constant once the input crossed its zener voltage, closely matching theory. Minor measurement variations were present, but the overall results were consistent and demonstrated effective voltage regulation.

Precautions:

- All devices were connected properly
- Avoid parallax error while taking readings from analog meters